

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT/CE)

May 2013

IT310 Artificial Intelligence (A.I.)

Date: 04.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions. 12

1. Explain search and problem space with the help of a suitable example.
2. Which is better when the memory is limited, Breadth First Search or Depth First Search? Give Reason(s).
3. What are the advantages and disadvantages of Hill Climbing Search Technique?
4. What are factors which decide the choice of reasoning forward or backward?
5. Compare Procedural Vs Declarative knowledge.
6. Discuss the important characteristics of production system and control strategy.

Q-2 Answer Any two from the following 10

- [A] What is meant by admissibility of heuristic search? Is A* algorithm admissible? Defend your answer.
- [B] What is production system? List the elements. Give some rules for water jug problem.
- [C] Convert these sentences to predicate logic. Using the logical rules, proof by resolution that prove "whether John eats Kumquats or not".
- (a) John likes fruit.
 - (b) All fruits are not liked by John.
 - (c) Kumquats are fruits.
 - (d) John eats only those fruits, which he likes.

Q-3 [A] Write a prolog program for the following. (Any two) 8

1. Reversing a list using difference list concept.
2. To add all elements of a given integer list.
3. For generating the subsists of the given list.

[B] Explain the principal of means end analysis approach to problem solving. 3

[C] Draw the block diagram of a typical Expert System. 2

OR

[C] Explain the term syntactic processing and semantic analysis in context to NLP. 2

SECTION-II

- Q-4** 12
1. What is a Clausal form? How it is related with PROLOG?
 2. How to decide about the training time required to train the neural network?
 3. What is meant by admissibility of heuristic search?
 4. Give three situations where cut and fail might be useful.
 5. Differentiate: Supervised and Unsupervised learning.
 6. Introduce monotonic reasoning with example.
- Q-5** 5
- [A] Compare Nonhierarchical Planning with Hierarchical Planning. 5
- OR**
- [A] Give an example of problems for which BFS would work better than DFS. Explain it in detail. 5
- [B] What is Backpropagation error? How is it used for batch training and incremental training? What learning rate should be used for back propagation? 5
- OR**
- [B] Draw the semantic net for following sentence: "Every dog in the town has bitten the constable". 5
- Q-6** 6
- [A] Justify the statement: Game tree is AND-OR Tree. 6
- OR**
- [A] Write grammar rules, which can handle the following sentence and also generate the parse tree. 6
RBI has raised the interest rates by 1%.
- [B] Explain the key idea of Bayesian statistics for representing knowledge. Is it possible to compute $P(A/\sim B)$? Explain. 7
- OR**
- [B] Trace the execution of the constraint satisfaction procedure in solving the crypt arithmetic problem: 7

$$\begin{array}{r}
 \text{SEND} \\
 + \\
 \text{MORE} \\
 \hline
 \text{MONEY}
 \end{array}$$